

-27/10/25

2. ALGEBRA EQUACIONES

Factorización de polinomios

- Sacar factor común

$$3x^4 - 6x^2 = \boxed{3x^2 \cdot (x^2 - 2)}$$

- Productos notables: $a^2 - b^2 = (a + b) \cdot (a - b)$

$$x^3 - x = x \cdot (x^2 - 1) = x \cdot (x + 1) \cdot (x - 1)$$

- $P = ax^2 + bx + c$ con $p \neq q \rightarrow P = (x - p) \cdot (x - q)$

$$6x^2 + 7x - 3 = 6 \cdot \left(x - \frac{1}{3}\right) \cdot \left(x - \frac{-3}{2}\right)$$

Resolver la ecuación
normal.

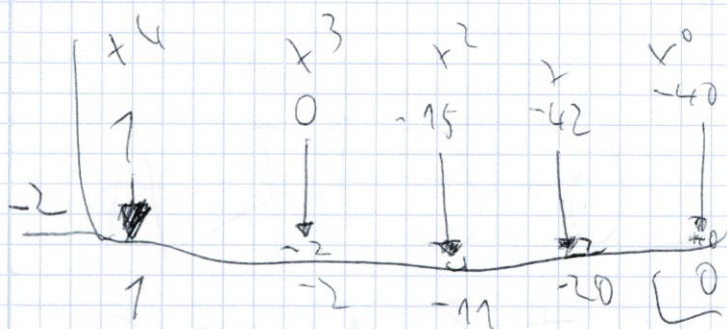
$$\frac{b \pm \sqrt{b^2 - 4ac}}{a} \cdot (x - \dots)$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{a} \begin{matrix} \swarrow \frac{1}{3} \\ \searrow \frac{-3}{2} \end{matrix}$$

División

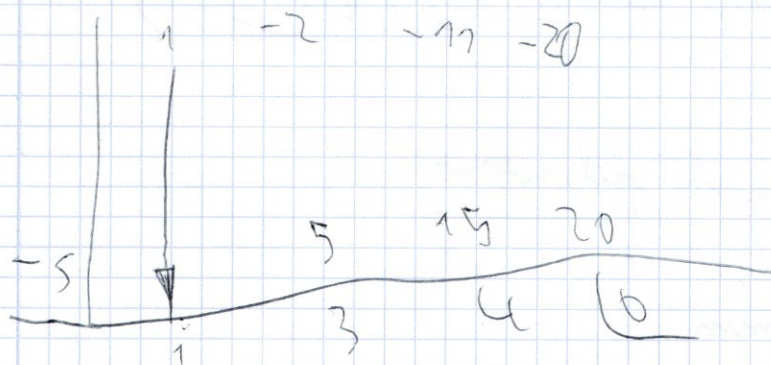
- Ruffini (ordenar x^2) de x^4

$$x^6 - 15x^4 - 42x^3 - 40x^2 = x^2 \cdot (x^4 - 15x^2 - 42x - 40)$$



$$= (x - (-2)) \cdot (x^3 - 2x^2 - 11x - 20)$$

$$= x + 2 \cdot (x^3 - 2x^2 - 11x - 20)$$



$$= x^2 \cdot (x + 2) \cdot (x - 5) \cdot (x^2 + 3x + 4)$$

Fracciones algebraicas

1. Factor común
2. Ruffini $\rightarrow$ $\pm n^2 - k^2$
 $\rightarrow$ Pares grados

$$\frac{ax^2}{x^2 - 3x} = \frac{(3+x)(3-x)}{x \cdot (x-3)} = (3+x) / (-1) = -3 - x$$

Opuesto	Normal
$\frac{(3-x)}{(-3+x)} = -1$	$\frac{x-3}{x-3} = 1$

Ejemplo 2.26
va a examen

Tipo de examen
2.28

$x = -1$ se substituye la x para comprobar raíz.
y el resultado sera $= 0$

Enda de repaso $\rightarrow 2, 3, 4, 11, 12$

① Sean los polinomios: $A(x) = 3x^2 + x - 4$, $P(x) = 2 - 3x^2$, $C(x) = 3x$,
 $R(x) = 3x^2 + 1$. Efectúa las operaciones indicadas a continuación.

a) $1(3x) - 3 \cdot (3x^2 + x - 4) + 3x = \cancel{9x^2} - \cancel{9x^2} - 3x + 12 + 3x = 12$

b) $(3x^2 + x - 4)^2 = 9x^4 + 6x^3 - 23x^2 - 8x + 16$

c) $(2 - 3x^2) \cdot (2 + 3x^2) + 9x^2 = -9x^4 + 9x^2 + 4$

d) $(3x^2 + 2)^3 = 27x^6 + 54x^4 + 36x^2 + 8$

② Que polinomio hay que restar a $A(x) = 4x^2 - x + 7$ para obtener $B(x) = x^2 - 1$.

$$4x^2 - x + 7 - 3x^2 - x + 8 = x^2 - 1$$

④ Si $A(x) = ax^2 + 3x - 4$, $B(x) = -x^2 + bx - c$,
 $C(x) = 3x^2 + 7x - 8$, que tienen que restar a, b y c para que $A(x) + B(x) = C(x)$.

$$ax^2 + 3x - 4 - x^2 + bx - c = 3x^2 + 7x - 8$$

$$ax^2 + 3x - 4 - x^2 + bx - c - 3x^2 - 7x + 8 = 0$$

$$ax^2 - 4x^2 - 4x + 4 = 0$$

$$bx - 4 = 0 \rightarrow b = 4$$

$$-c + 4 = 0 \rightarrow c = 4$$

11) Comprobar si $x = -1$ es raíz del polinomio $x = -1 \pm 3$
 Raíz del polinomio $A(x) = -2x^2 + 3x + 5$.

$$-2 \cdot (-1)^2 + 3 \cdot (-1) + 5$$

$$-2 - 3 + 5 = 0 \quad \text{Si.}$$

12) Comprobar si $x = -2$ es raíz del polinomio $A(x) =$
 $-x^3 - 2x^2 - 3x - 6$

$$-(-2)^3 - 2 \cdot (-2)^2 - 3 \cdot (-2) - 6 = 0 \quad \text{Si.}$$

20/10/25

26) Trabaja usando el método de Ruffini.

$$a) 2x^3 + 4x^2 - 2x - 4 =$$

$$(x-1) \cdot (2x^2 + 6x + 4)$$

$\begin{array}{r rrrr} & 2 & 6 & 4 & \\ \hline -2 & & -4 & -4 & \\ \hline & 2 & 2 & 0 & \end{array}$	$\begin{array}{r rrrrr} & 2 & 4 & -2 & -4 & \\ \hline 1 & & 2 & 6 & 4 & \\ \hline & 2 & 6 & 4 & 0 & \end{array}$
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$(x-1) \cdot (x+2) \cdot (2x-2)$

$$(x-1) \cdot (x+2) \cdot (x+1) R=2$$

$\begin{array}{r rr} & 2 & 2 \\ \hline -1 & & -2 \\ \hline & 2 & 0 \end{array}$

$$b) 3x^3 + 4x^2 - 12x$$

$\begin{array}{r rrrrr} & 3 & 4 & -12 & 0 & \\ \hline 1 & & 3 & 12 & 0 & \\ \hline & 3 & 12 & 0 & 0 & \end{array}$	$(x-1) \cdot (x+4) R=3$
$\begin{array}{r rrrr} & 3 & 4 & -12 & 0 \\ \hline -4 & & -12 & -12 & 0 \\ \hline & 3 & 0 & 0 & 0 \end{array}$	

27) Factoriza usando identidades notables.

$$a) 4x^2 - 1 = (2x+1)(2x-1)$$

$$b) 9x^2 - 25 = (3x-5)(3x+5)$$

$$c) x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2})$$

$$d) 4x^2 = (3x+2) \cdot (-3x+2)$$

$$e) x^2 - 6x + 9 = (x-3)^2$$

$$f) x^4 - 5x^2 + 4 = (x+4)^2$$

$$g) x^2 - 1 = (\sqrt{2}x+1) \cdot (\sqrt{2}x-1)$$

$$h) 4x^2 + 4x + 1 = (2x+1)^2$$

$$i) 9x^2 - 12x + 4 = (3x-2)^2$$

$$j) x^2 - 6xy + 9y^2 = (x-3y)^2$$

34) simplifica las siguientes fracciones.

$$a) \frac{x^2 + 25 - 10x}{x^2 - 25} = \frac{(x-5)^2}{(x-5)(x+5)} = \frac{(x-5) \cdot \cancel{(x-5)}}{\cancel{(x-5)} \cdot (x+5)}$$

$$b) \frac{(x-3)^2 \cdot (x+3)}{(x-3) \cdot x^2 \cdot (x+2)} = \frac{(x-3) \cdot (x+3)}{x \cdot (x+2)}$$

$$c) \frac{x^2 - 1}{x - 1} = \frac{\cancel{(x-1)} \cdot (x+1)}{\cancel{(x-1)}} = x+1$$

$$d) \frac{(x-2)}{(x^2 + x - 6)} = \frac{\cancel{(x-2)}}{(x+3)\cancel{(x-2)}} = \frac{1}{(x+3)}$$

$$e) \frac{x^3 + 3x^2 + x + 3}{x^3 + 3x^2} = \frac{x^2 \cdot (x+3) + 1 \cdot \cancel{(x+3)}}{x^2 \cdot \cancel{(x+3)}} = \frac{x^2 + 1}{x^2}$$

$$\begin{aligned} & \frac{1}{x^3-5x^2+6x} = \frac{x \cdot (x^2-5x+6)}{x \cdot (x-2)(x-3)} \\ & \frac{x^3-5x^2+6x}{x^3-x^2-14x+24} = \frac{(x-2)(x+4)(x-3)}{(x+4)} \\ & = \frac{x}{x+4} \end{aligned}$$

$$\begin{array}{r|rrrr} & 1 & -1 & -14 & +24 \\ 2 & & 2 & 2 & -24 \\ \hline & 1 & 1 & -12 & 0 \\ -4 & & 4 & +12 & \\ \hline & 1 & 3 & 0 & \\ & & 3 & & \\ \hline & 1 & 0 & & \end{array}$$

③ Reduce to common denominator & solve

$$a) \frac{x}{x-3} - \frac{3}{x} = \frac{x^2}{x(x-3)} - \frac{3 \cdot (x-3)}{x \cdot (x-3)} = \frac{x^2 - 3x + 9}{x \cdot (x-3)}$$

$$b) \frac{2}{x-2} + \frac{2}{x+2} = \frac{(x-2)2 + 2 \cdot (x+2)}{(x-2)(x+2)} = \frac{2x-4+2x+4}{(x-2)(x+2)} = \frac{4x}{(x-2)(x+2)}$$

$$\frac{4x}{(x-2)(x+2)}$$

$$c) \frac{2}{x^2} - \frac{x+1}{3x} = \frac{6}{3x^2} - \frac{x^2+x}{3x^2} = \frac{6-(x^2+x)}{3x^2} = \frac{6-x^2-x}{3x^2}$$

$$\frac{-x^2-x+6}{3x^2}$$

$$c) 2x^4 - 20x^2 + 18$$

$$(x+3) \cdot (x+1) \cdot (x-1) \cdot (x-3)$$

$$R=2$$

$$\begin{array}{r|rrrrrr} & 2 & 0 & -20 & 0 & +18 & \\ -3 & \downarrow & & -6 & -18 & +6 & -18 \\ \hline & 2 & & -6 & -2 & +6 & \boxed{0} \\ -1 & \downarrow & & -2 & +8 & -6 & \\ \hline & 2 & & -8 & 6 & & \boxed{0} \\ 1 & \downarrow & & 2 & & -6 & \\ \hline & 2 & & -6 & & & \boxed{0} \\ 3 & \downarrow & & 6 & & & \\ \hline & 2 & & & & & \boxed{0} \end{array}$$

$$d) x^2 - 7x + 6$$

$$(x-1) \cdot (x-6)$$

$$R=1$$

$$\begin{array}{r|rrr} & 1 & -7 & 6 \\ +1 & \downarrow & 1 & -6 \\ \hline & 1 & -6 & \boxed{0} \\ 6 & \downarrow & 6 & \\ \hline & 1 & & \boxed{0} \end{array}$$

$$e) x^3 + 3x^2 - x - 3$$

$$(x+3) \cdot (x+1) \cdot (x-1) =$$

$$R=-9$$

$$\begin{array}{r|rrrr} & 1 & 3 & -1 & -3 \\ -3 & \downarrow & -3 & 0 & 3 \\ \hline & 1 & 0 & -1 & \boxed{0} \\ -1 & \downarrow & -1 & & 1 \\ \hline & 1 & -1 & & \boxed{0} \\ 1 & \downarrow & 1 & & \\ \hline & 1 & & & \boxed{0} \end{array}$$

$$f | x^4 - 5x^2 + 4 =$$

$$(x+2) \cdot (x+1)(x-1)(x-2)$$

$$R=1$$

$$\begin{array}{r|rrrrr} & 1 & 0 & -5 & 0 & 4 \\ -2 & \downarrow & -2 & +4 & +2 & -4 \\ \hline & 1 & -2 & -1 & +2 & 0 \\ -1 & \downarrow & -1 & +3 & -2 & \\ \hline & 1 & -3 & 2 & 0 & \\ 2 & \downarrow & 2 & & & \\ \hline & 1 & & & & 0 \end{array}$$

$$g | -2x^3 + x^2 + 8x - 4 = -2(x+2)(x-2) \cdot (x - \frac{1}{2})$$

$$\begin{array}{r|rrrr} & -2 & 1 & 8 & -4 \\ -2 & \downarrow & +4 & -10 & +4 \\ \hline & -2 & +5 & -2 & 0 \\ 2 & \downarrow & -4 & & \\ \hline & -2 & 1 & 0 & \\ \frac{1}{2} & \downarrow & -1 & & \\ \hline & -2 & 0 & & \end{array}$$

$$h | -5x^3 + 3x^2 + 20x - 12$$

$$-5(x+2)(x-2)(x - \frac{3}{5})$$

$$\begin{array}{r|rrrr} & -5 & 3 & 20 & -12 \\ -2 & \downarrow & +10 & -26 & +2 \\ \hline & -5 & 13 & -6 & 0 \\ 2 & \downarrow & -10 & 6 & \\ \hline & -5 & 3 & 0 & \\ \frac{3}{5} & \downarrow & -3 & & \\ \hline & -5 & 0 & & \end{array}$$

-31/10/25

Repaso 3, 13, 14, 32

3) Opera y simplifica cuando sea posible.

$$4) \frac{5x-10}{x+3} \cdot \frac{x^2-4}{x-2} = \frac{5x-10}{x+3} \cdot \frac{x^2-3^2}{x-2} = \frac{5 \cdot (x-2) \cdot (x-3) \cdot (x+3)}{(x-2) \cdot (x+3)} = 5(x-3)$$

$$6) \frac{2x+1}{2x-1} \cdot \frac{x^2}{4x-2} = \frac{(2x+1) \cdot (4x-2)}{(2x-1) \cdot x^2} = \frac{(2x+1) \cdot 2 \cdot (x-1)}{x^2 \cdot (2x-1)} = \frac{2(2x+1)}{x^2}$$

$$1) \frac{x}{3} \cdot \frac{2x+1}{x-1} = \frac{2x^2+x}{3x-3}$$

$$d) \frac{1}{x-1} \cdot \frac{x+1}{3x} = \frac{1}{x-1} \cdot \frac{3x}{x+1} = \frac{3x}{(x-1) \cdot (x+1)} = \frac{3x}{x^2-1}$$

$$\boxed{\begin{array}{l} x^2-1 = (x+1)(x-1) \\ (x-0)(x-0)(x+1) \\ 2 \cdot (x-1) \\ x^3+x^2 \end{array}}$$

$$2) \frac{2}{x-1} \div \frac{x}{x+1} = \frac{2(x+1)}{x^2-x} = \frac{2x+2}{x^2-x}$$

$$8) \frac{2x}{2x-3} \div \frac{x+1}{2x-3} = \frac{2x}{2x+3} \cdot \frac{2x+3}{x+1} = \frac{4x^2+6x}{2x^3+3x+3+2x}$$

$$\frac{4x^2+6x}{2x^3-x+3}$$

13) Escribe en cada apartado un polinomio de coeficientes enteros con la raíz indicada.

$$a) 7y-7$$

$$(x-7) = x^2 - 49$$

$$(x+7)$$

$$b) -2y-3$$

$$(x+2)$$

$$(x+3) = x^2 + 5x + 6$$

$$c)$$

$$0 \ 0 \ -1$$

$$(x-0) \quad x^2 \cdot (x+1) = x^3 + x^2$$

$$(x-0)$$

$$(x+1)$$

$$d)$$

$$2 \ 2 \ -3 \ -3$$

$$(x-2)$$

$$(x-2) = x^2 - 4x + 4$$

$$x^4 + 2x^3 - 11x^2 - 12x + 36$$

$$(x+3) = x^2 + 6x + 9$$

$$(x+3)$$

14) Escriba en cada apartado un polinomio de coeficientes enteros con la raíz indicada.

$$a) x_1 = 4 \ x_2 = -2 \ x_3 = 0 \quad (\text{triple})$$

$$x_1 \rightarrow (x-4)$$

$$x_2 \rightarrow (x+2)$$

$$x_3 \quad (x-0) \quad x^2 \cdot (x+2) \cdot (x-4) = x^2 \cdot (x^2 - 2x - 8)$$

$$x_3 \rightarrow (x-0) \quad x^4 - 2x^3 - 8x^2$$

b)

$$x_1 = 4 \quad (x-4) \quad \rightarrow \quad x^2 - 2x - 8$$

$$x_2 = -2 \quad (x+2)$$

$$x_3 = -3 \quad (x+3)$$

$$x_4 = 0 \quad (x)$$

$$x^2 + 3x \rightarrow x \cdot (x+3)$$

$$x^4 + x^3 - 14x^2 - 24x$$

c)

$$x_1 = -\frac{1}{4} \rightarrow (x + \frac{1}{4})$$

$$x_2 = 1 \rightarrow (x+1)$$

$$x_3 = -1 \rightarrow (x-1)$$

$$x_4 = -1 \rightarrow (x+1)$$

$$x^2 + \frac{5}{4}x + \frac{1}{4}$$

$$x^2 + 2x + 1$$

$$= x^4 + 13x^3 + 15x^2 + 7x + 1$$

d)

$$x_1 (3x-2)$$

$$x_2 (x+2)$$

$$x_3 (3x+1)$$

$$x_4 x$$

$$3x^2 + 4x - 4 \cdot 3x + 1 = 9x^2 + 15x^2 - 8x - 4$$

32) Escribe en cada grado un polinomio con la característica indicada.

a) de segundo grado irreducible: $(x^2 + 1)$

b) de segundo grado con dos raíces reales: $(x - 1)^2$

c) de tercer grado irreducible: $(x^3 - (x - 1))$

d) de cuarto grado irreducible: $(x^4 + 1)$

e) de tercer grado con una raíz simple: $(x^3 - x^2 + x - 1)$

f) de tercer grado con una raíz triple: $(x^3 + 3x^2 + 3x + 1)$

-3/11/25

EJERCICIOS

Fichas de referencia: 62, 63, 77, 78

$$\sqrt{2x-3} + \sqrt{x+7} = 4$$

$$(\sqrt{2x-3})^2 = (4 - \sqrt{x+7})^2$$

$$2x-3 = 16 + x+7 - 8\sqrt{x+7}$$

$$(x-26)^2 = (8\sqrt{x+7})^2$$

$$x^2 + 26^2 - 52x = 64 \cdot (x+7)$$

$$x^2 - 116x + 26^2 - 64 \cdot 7 = 0$$

$$a) 3^{1-x^2} = \frac{1}{27}$$

$$3^{1-x^2} = \frac{1}{3^3}$$

$$3^{1-x^2} = 3^{-3}$$

$$-x^2 = -4$$

$$x^2 = 4$$

$$x = \pm \sqrt{4} = \pm 2$$

$$b) 5x^2 - 5x + 6 = 1$$

$$5x^2 - 5x + 6 = 5^0$$

$$x^2 - 5x + 6 = 0$$

$$(x-2)(x-3) = 0$$

$$x = 2$$

$$x = 3$$

c)

$$2^x + 2^{x+1} = 12$$

$$2^{x+1} = 2^x \cdot 2^1$$

$$2 \cdot 2^x + 2^x = 12$$

$$2^x \cdot (2+1) = 12$$

$$2^x \cdot (3) = 12$$

$$x = \frac{12}{3}$$

$$2^x = 4$$

$$x = \frac{4}{2} = 2$$

	1	-5	+6
+2	1	2	-6
	1	-3	0
3	1	3	0
	1	0	0

$$d) 2^{3x-1} - 4^{x+3} = 0$$

$$2^{3x-1} = 4^{x+3} \quad 2^{3x-1} = (2^2)^{x+3}$$

$$2^{3x-1} = 2^{2x+6}$$

$$3x-1 = 2x+6$$

$$x - -7 = 0$$

$$x = -7$$

$$e) 3^x + \frac{1}{3^{x+1}} = \frac{28}{9}$$

$$3^x + \frac{1}{3^{x+1}} = \frac{28}{9}$$

$$3^x + \frac{1}{3^{x+1}} = \frac{28}{9}$$

$$3^x + \frac{1}{3^x} + \frac{1}{3} = \frac{28}{9}$$

$$3^x + \frac{1}{3^x} = \frac{28}{9}$$

$$y = 3^x$$

$$y + \frac{1}{y} = \frac{28}{9}$$

$$9y(y) + 9y \left(\frac{1}{3y} \right) = 9y \left(\frac{28}{9} \right)$$

$$9y^2 + 3 = 28y$$

$$ay^2 + 3 - 28y = 0$$

$$ay^2 - 28y + 3 = 0$$

$$D = b^2 - 4 \cdot a \cdot c = (-28)^2 - 4 \cdot a \cdot 3 = 616$$

$$y = \frac{+28 \pm 26}{18}$$

$$3^x = 3^1 \quad x = 1$$

$$3^x = \frac{1}{3} \quad 3^x = 3^{-2} \quad x = -2$$

$$f) 2^{x+1} + 2^{x-1} = \frac{5}{2}$$

$$2^x \cdot 2 + \frac{2^x}{2} = \frac{5}{2}$$

$$2^x \left(\frac{4}{2} + \frac{1}{2} \right) = \frac{5}{2}$$

$$2^x \cdot \frac{5}{2} = \frac{5}{2}$$

$$2^x = \frac{5}{2} : \frac{5}{2}$$

$$2^x = 1$$

$$2^0 = 1$$

$$\boxed{x = 0}$$

$$g) 8^{1+4x} + 2^{3x-1} = \frac{17}{16}$$

$$2^{3+3x} + 2^{3x-1} = \frac{17}{16}$$

$$2^3 \cdot 2^{3x} + \frac{2^{3x}}{2^1} = \frac{17}{16}$$

$$8 \cdot 2^{3x} + \frac{2^{3x}}{2} = \frac{17}{16}$$

$$y = 2^{3x}$$

$$8y + \frac{y}{2} = \frac{17}{10}$$

$$128y + 80y = 17$$

$$136y = 17$$

$$y = \frac{17}{136} = \frac{1}{8}$$

$$\frac{1}{8} = 2^{3x}$$

$$2^{-3} = 2^{3x}$$

$$-3 = 3x \quad x = -\frac{1}{3}$$

$$h | 2^x - 5 \cdot 2^x + 4 = 0$$

$$(2^x)^2 - 5 \cdot 2^x + 4 = 0 \quad y = 2^x$$

$$y^2 - 5y + 4 = 0$$

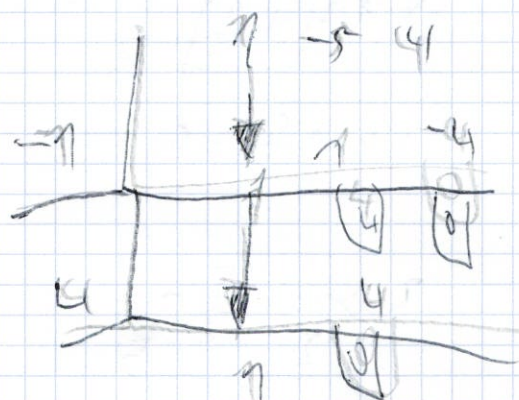
$$(y-1) \cdot (y-4)$$

$$y = 1$$

$$y = 4$$

$$1 = 2^x \quad x = 0$$

$$4 = 2^x \quad x = 2$$



$$114^x - 3^x = 6$$

$$(3^{x+1})^2 - 3^x = 6$$

$$y = 3^x$$

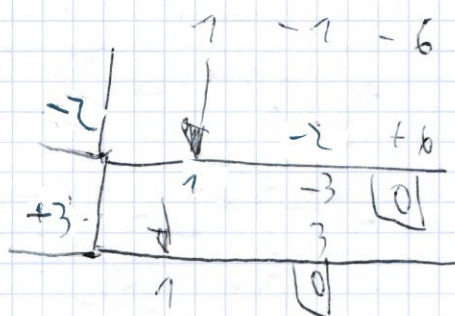
$$y^2 - y - 6 = 0$$

$$y = 2$$

$$y = -3$$

$$3^x = 2 \quad \text{NO EXISTE}$$

$$3^x = -3 \quad x = -1$$



-5/11/25

$$5x^2 - 5x + 6 = 1$$

$$\ln(5x^2 - 5x + 6) = \ln 1$$

$$x^2 - 5x + 6 = \ln 1$$

$$x^2 - 5x + 6 = 0$$

$$\Delta = (-5)^2 - 4 \cdot 1 \cdot 6$$

$$\Delta = +25 - 24$$

$$\Delta = 1$$

$$x = \frac{-5 \pm 1}{2} \quad \begin{matrix} \frac{6}{2} = 3 \\ \frac{4}{2} = 2 \end{matrix}$$

$$3^x + 1 = \frac{28}{3^x + 1}$$

$$\frac{3^x + 1}{1} = \frac{28}{3^x + 1}$$

$$3^x + 1$$

$$\frac{3^x \cdot 3^{x+3}}{3^{x+3}} + \frac{3^2}{3^{x+3}} = \frac{28 \cdot 3^{x+1}}{3^{x+3}}$$

$$\text{MLM} = 3^{x+1} \cdot 3^2 = 3^{2x+3}$$

$$3^x \cdot 3^{x+3} + 3^2 = 28 \cdot 3^{x+1}$$

$$3^{2x+3} + 3^2 = 28 \cdot 3^{x+1}$$

63) Resolver las siguientes ecuaciones con fracciones algebraicas.

$$a) \frac{6}{x} + \frac{x+1}{x-2} = 6 \quad \frac{6x-12+x^2+x}{x^2-2x} = \frac{x^2+4x-12}{x^2-2x} = 6$$

$$x^2+4x-12=6x^2-12x$$

$$x^2-6x^2+4x+12x-12=0$$

$$-5x^2+16x-12=0$$

$$a=-5 \quad b=16 \quad c=-12$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-16 \pm 16}{-10} \quad \begin{matrix} \swarrow \frac{4}{5} \checkmark \\ \searrow 3 \checkmark \end{matrix}$$

$$b) \frac{2x-3}{x^2-5x} = \frac{x+4}{x} - \frac{3}{4} \rightarrow 4x(x-5)$$

$$\frac{8x-12}{4x(x-5)} + \frac{4x^2-4x-80}{4x(x-5)} = \frac{3x^2-15x}{4x(x-5)}$$

$$8x-12+4x^2-4x-80=3x^2-15x$$

$$-x^2+14x-92=0$$

$$a=-1 \quad b=14 \quad c=-92$$

$$D=117$$

$$x = \frac{-14 \pm \sqrt{117}}{2} \quad \begin{matrix} \swarrow 23 \checkmark \\ \searrow 1 \checkmark \end{matrix}$$

$$1) \frac{2x^3}{x} - \frac{3}{x^2} = -\frac{1}{1}$$

$$\frac{2x^2}{x^2} - \frac{3}{x^2} = -\frac{1}{x^2}$$

$$\frac{2x^4 - 3 + x^2}{x^2} = 0$$

$$2x^4 + x^2 - 3 = 0$$

$$y = x^2 \quad 2y^2 + y - 3 = 0 \quad \begin{matrix} 1 \\ 3 \\ 2 \end{matrix} \begin{matrix} \vee \\ \times \\ \times \end{matrix}$$

$$2) \frac{x+3}{x-1} - \frac{x^2+1}{x^2-1} = \frac{26}{35}$$

$$\frac{x+3}{x-1} - \frac{x^2+1}{(x-1)(x+1)} = \frac{26}{35}$$

$$\frac{x^2+3x+4x-x^2-1}{(x-1)(x+1)} \rightarrow \frac{x^2-x^2+4x+2}{(x-1)(x+1)} = \frac{4x+2}{(x-1)(x+1)} = \frac{26}{35}$$

$$\frac{4x+2}{x^2-1} = \frac{26}{35} \quad 26x^2 - 26 = 140x + 42$$

$$26x^2 - 140x - 68 = 0 \quad 13x^2 - 70x - 34 = 0$$

$$A = b^2 - 4 \cdot a \cdot c = 70^2 + 396$$

$$x = \frac{70 \pm 86}{26} = \begin{matrix} 2 \\ -6 \end{matrix}$$

$$2) \sqrt{3x + \frac{1}{x}} - \frac{1}{x} = 1$$

$$\left(\frac{1}{x} + 1\right)^2 = \left(\sqrt{3x + \frac{1}{x}}\right)^2$$

$$x^2 \left(1 + \frac{2}{x} + \frac{1}{x^2}\right) = \left(3x + \frac{1}{x}\right)$$

$$x^2 + 2x + 1 = 3x^3 + x$$

$$3x^3 - x^2 - x - 1 = 0$$

$$\cancel{(x+1)} \quad (x-1) \rightarrow 3 - 1 - 1 - 1 = 0$$

$$\begin{array}{r|rrrr} & 3 & -1 & -1 & -1 \\ 1 & 3 & 2 & 1 & 0 \end{array}$$

$$(x-1) \quad \boxed{x=1}$$

$$3x^2 + 2x + 1 = 0$$

$$\Delta = -8 \rightarrow 0 \text{ solution}$$

$$1) 2\sqrt{x-1} + \frac{x+4}{x} = 5$$

$$2\sqrt{x-1} = 5 - \frac{x+4}{x}$$

$$2\sqrt{x-1} = \frac{5x - (x+4)}{x}$$

$$2\sqrt{x-1} = \frac{5x - x - 4}{x}$$

$$(2\sqrt{x-1})^2 = \left(\frac{4x-4}{x}\right)^2$$

$$4 \cdot (x-1) = \frac{(4 \cdot (x-1))^2}{x^2}$$

$$4 \cdot (x-1) = \frac{16 \cdot (x-1)^2}{x^2}$$

$$4x^2 \cdot (x-1) = 16 \cdot (x-1)^2$$

$$4x^2 = 16 \cdot (x-1)$$

$$x-1 \sqrt{\frac{16}{4}} = 12$$

$$x-1$$

ⓧ Remember to simplify fractions exponentially.

$$d) \sqrt{2x-3} + 1 = x \rightarrow 2x-3 = (x-1)^2 \Rightarrow 2x-3 = x^2 - 2x + 1$$

$$x^2 - 4x + 4 = 0$$

$$x=2$$

$$x = \frac{4 \pm \sqrt{16 - 4 \cdot 4}}{2} = \frac{4}{2} = 2$$

$$k) \sqrt{2x-3} + \sqrt{x+7} = 4 \rightarrow \sqrt{2x-3} = 4 - \sqrt{x+7}$$

$$2x-3 = 16 - 8\sqrt{x+7} + (x+7) \quad 2x-3+1+8\sqrt{x+7}$$

$$2x-3-x-23 = -8\sqrt{x+7} \quad x-26-8\sqrt{x+7}$$

$$(x-26)^2 = (-8\sqrt{x+7})^2$$

$$x^2 - 52x + 676 = 64(x+7)$$

$$x^2 - 52x + 676 - 64x - 448 = 0$$

$$x^2 - 116x + 228 = 0$$

$$x = \frac{116 \pm \sqrt{116^2 - 4 \cdot 228}}{2} = \frac{116 \pm 112}{2} \quad \begin{matrix} \swarrow \sqrt{144} \\ \searrow \end{matrix}$$

e)

$$\sqrt{2\sqrt{x+2}} = 4$$

$$2\sqrt{x+2} = 16$$

$$\sqrt{x+2} = 8$$

$$x+2 = 64$$

$$x = 64 - 2 = 62$$

d)

$$x+1 + \sqrt{2x^2+2x-3} = 0$$

$$\sqrt{2x^2+2x-3} = -x-1$$

$$2x^2+2x-3 = (-x-1)^2 = x^2+2x+1$$

$$2x^2-x^2+2x-2x-3-1=0 \quad x^2-4=0 \quad x=\pm\sqrt{4}$$

$$x=2 \rightarrow -2$$

$$x=2 \rightarrow -4$$

$$2) \sqrt{1-4x} - 2\sqrt{3+x} = \sqrt{3+x}$$

$$\sqrt{1-4x} = 3\sqrt{3+x}$$

$$1-4x = 9(3+x)$$

$$-1-4x = 27+9x$$

$$-13x = 26$$

$$x = \frac{26}{-13} = -2$$

$$f) 2x^2 - 3\sqrt{x^2-3} - 5 = 0$$

$$\left(\frac{2x^2-5}{3} \right)^2 = (x^2-3)$$

$$\frac{(2x^2-5)^2}{9} = x^2-3$$

$$(2x^2-5) \cdot (x^2-2) = 0$$

$$4x^4 - 20x^2 + 25 = 0$$

$$4x^4 - 20x^2 + 25 = 0$$

$$y = x^2$$

$$4y^2 - 20y + 25 = 0$$

$$y = \frac{10 \pm \sqrt{10^2 - 4 \cdot 4 \cdot 25}}{8}$$

$$\frac{10 \pm 3}{8}$$

$$4 = x^2$$

$$x = \sqrt{4} = 2$$

$$3/25 = x^2 \quad x = \sqrt{3/25} = 1/5$$

$$g) 7^{1+2x} - 50 \cdot 7^x + 7 = 0$$

$$7 \cdot (7^x)^2 - 50 \cdot 7^x + 7 = 0$$

$$y = 7^x$$

$$7y^2 - 50y + 7 = 0$$

$$7y^2 - 50y + 7 = 0$$

$$a = 7 \quad b = -50 \quad c = 7$$

$$\Delta = b^2 - 4 \cdot a \cdot c = (-50)^2 - 4 \cdot 7 \cdot 7 = 2304$$

$$x = \frac{-50 \pm 48}{14}$$

$$\begin{array}{l} 7 \\ 1 \\ 1 \\ 7 \end{array}$$

$$7^x = 7 \quad x = 1$$

$$7^x = \frac{1}{7} \quad x = -1$$

7.8) Remember logarithmic equations logarithms.

$$a) \log x + \log 50 = 3$$

$$\log(50x) = 3$$

$$\log a^b = c \quad a^b = c$$

$$10^3 = 50x$$

$$x = \frac{1000}{50} = 20$$

$$x = 20$$

$$\text{le } 15 \log_2(x+3) = \log_2 32$$

$$\log_2(x+3)^5 = \log_2 32$$

$$(x+3)^5 = 32$$

$$(x+3) = 2$$

$$x = 2 - 3$$

$$x = -1$$

$$(1) 2 \log x = \log(10-3x)$$

$$\log x^2 = \log(10-3x)$$

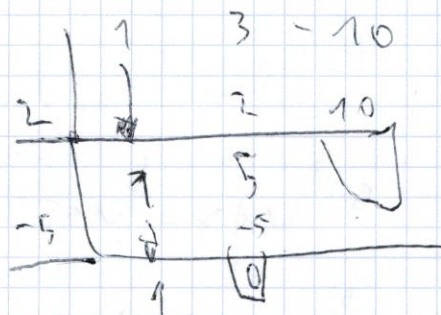
$$x^2 = 10-3x$$

$$x^2 + 3x - 10 = 0$$

$$(x-2)(x+5)$$

$$x = -5 \text{ NO}$$

$$x = 2$$



$$d) \log(5x-3) = \frac{4}{5}$$

$$10^{\frac{4}{5}} = 5x-3$$

$$x = \frac{10^{\frac{4}{5}} + 3}{5}$$

$$x = \frac{\sqrt[5]{10^4} + 3}{5}$$

$$x = \frac{\sqrt[5]{10^4}}{\sqrt[5]{5}} + \frac{3}{5}$$

$$2) \frac{1}{2} \log(2x+3) = \log x$$

$$\log(2x+3)^{\frac{1}{2}} = \log x$$

$$\log \sqrt{2x+3} = \log x$$

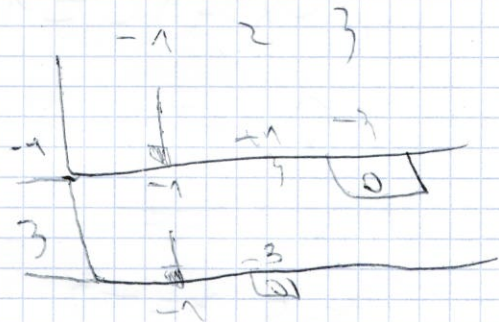
$$(\sqrt{2x+3})^2 = (x)^2$$

$$-x^2 + 2x + 3 = 0$$

$$-(x+1)(x-3)$$

$$x = 3$$

$$x = -1 \text{ no}$$



$$f) \ln(x-1) + \ln(x+6) = \ln(3x+2)$$

$$\ln(x-1) \cdot (x+6) = \ln(3x+2)$$

$$(x-1) \cdot (x+6) = (3x+2)$$

$$x^2 + 5x - 6 = 3x + 2$$

$$x^2 + 2x - 8 = 0$$

$$(x-4)(x+2)$$

$$x = -4 \quad \vee \quad 0$$

$$x = -2$$

$$g) \log(x^2+1) - \log(x^2-1) = \log \frac{13}{12}$$

$$\log \frac{x^2+1}{x^2-1} = \log \frac{13}{12}$$

$$\frac{x^2+1}{x^2-1} = \frac{13}{12}$$

$$12x^2 + 12 = 13x^2 - 13$$

$$12 \cdot 5 = x^2$$

$$x = \pm \sqrt{25}$$

$$x = 5 \quad x = -5$$

$$4) \log(x+9) = 2 + \log x$$

$$\log(x+9) - \log x = 2$$

$$\log \frac{x+9}{x} = 2$$

$$\frac{x+9}{x} = 10^2$$

$$\frac{x+9}{x} = 100$$

$$x+9 = 100x$$

$$9 = 100x - x = 99x \quad 9 = 99x \quad x = \frac{9}{99} \quad x = \frac{1}{11}$$

$$x = \frac{1}{11}$$

$$5) \log \sqrt{3x+5} + \log \sqrt{x} = 1$$

$$\log \sqrt{3x+5} \cdot \sqrt{x} = 1$$

$$\log \sqrt{3x^2+5x} = 1$$

$$\sqrt{3x^2+5x} = 10^1$$

$$(\sqrt{3x^2+5x}) = (10^1)^2$$

$$3x^2+5x = 10^2$$

$$3x^2 + 5x - 100 = 0$$

$$\Delta = b^2 - 4 \cdot a \cdot c = 5^2 - 4 \cdot 3 \cdot (-100) = 1225$$

$$x = \frac{-5 \pm 35}{6} \Rightarrow \frac{30}{6} = 5 \quad x = 5$$

$$\frac{-40}{6} = -\frac{20}{3} \approx 0$$

$$2(\log x)^2 + 7 \log x - 9 = 0$$

$$y = \log x$$

$$2y^2 + 7y - 9 = 0$$

$$\Delta = b^2 - 4 \cdot a \cdot c = 7^2 - 4 \cdot 2 \cdot (-9) = 121$$

$$y = \frac{-7 \pm 11}{4} = \begin{matrix} 1 \\ 1 - \frac{9}{2} \end{matrix} \quad \begin{matrix} 1 = \log x \\ 1 - \frac{9}{2} = \log x \end{matrix}$$

$$x_1 = 10^1$$

$$x_2 = 10^{-\frac{9}{2}}$$

40,46

40) Résolvez les systèmes inéquations polynomiales

a) $1 - 2(x - 2) \geq 3 - 5x$

$$1 - 2x + 4 \geq 3 - 5x$$

$$5 - 2x \geq 3 - 5x$$

$$3x \geq -2$$

$$x \geq \frac{-2}{3}$$

$$S = \left(\frac{-2}{3}, +\infty \right)$$

b) $3x - 4 < -7$

$$3x < -3$$

$$x < \frac{-3}{3}$$

$$x < -1$$

$$S = (-\infty, -1)$$

c) $\frac{1-2x}{5} < 3 - \frac{x}{2} \rightarrow \frac{2-4x}{10} < \frac{30}{10} - \frac{5x}{10} \rightarrow 2(1-2x) < 30-5x$

$$2 - 30 < -5x + 4x$$

$$-28 < -1x$$

$$(-\infty, 28)$$

$$28 > 1x \quad x < 28$$

$$d) x^2 - 5x + 4 > 0$$

$$(x-4)(x+1) > 0$$

$$x^2 - 5x + 4$$

$-\infty, -1$	1	$(1, 4)$	4	$(4, +\infty)$
	$\uparrow$	$\uparrow$		$\uparrow$

$$x \in (-\infty, -1)$$

$$x \in (1, 4)$$

$$x \in (4, +\infty)$$

$$(-\infty, -1) \cup (4, +\infty)$$

$$e) x^2 - 4x + 3 \leq 0$$

$$(x-1)(x-3) \leq 0$$

$$x^2 - 4x + 3$$

$(-\infty, 1)$	1	$(1, 3)$	3	$(3, +\infty)$
	$\uparrow$	$\uparrow$		$\uparrow$

$$[1, 3]$$

$$f) x^2 - 8 < 2x$$

$$x^2 - 2x - 8 < 0$$

$$(x-4)(x+2) < 0$$

$$-\infty, -2$$

-2	-2	4	$(4, +\infty)$
	$\uparrow$		

$$x^2 - 2x - 8$$

$$[-2, 4]$$

$$g) -x^2 - 3x > 2$$

$$-x^2 - 3x - 2 > 0$$

$$+x^2 + 3x + 2 < 0$$

$$(x+2)(x+1) < 0$$

$$(-\infty, -2) \cup (-2, -1) \cup (-1, +\infty)$$

$$(-2, -1)$$

$$h) \frac{2x^2}{2} \leq \frac{x-3}{2}$$

$$2x^2 \leq x-3$$

$$2x^2 - x + 3 \leq 0$$

$$x = \frac{1 \pm 0}{4}$$

Sim Solution

$$i) 3x^2 + 5 < 0$$

Simple

$$j) \frac{2x^2 + 1}{3} < \frac{-3x}{3}$$

$$2x^2 + 1 < -3x$$

$$2x^2 + 3x + 1 < 0$$

$$x = \frac{-3 \pm 1}{4} = \frac{-2}{4} = -\frac{1}{2}$$

$$(-\infty, -1) \cup (-1, -\frac{1}{2}) \cup (-\frac{1}{2}, \infty)$$

$$(-1, -\frac{1}{2})$$

$$k) -4x^2 - 3x > -1$$

$$-4x^2 - 3x + 1 > 0$$

$$x = \frac{3 \pm 5}{-8} = -\frac{1}{4}$$

$$(-\infty, -1) \cup (-1, \frac{1}{4}) \cup (\frac{1}{4}, \infty)$$

$$(-1, \frac{1}{4})$$

$$l) 6x^2 \leq 5x + 21$$

$$6x^2 - 5x - 21 \leq 0$$

$$x = \frac{-5 \pm 23}{12} = \frac{3}{2}$$

$$(-\frac{3}{2}, \frac{7}{3})$$

m1

$$(1-\sqrt{3}, 1+\sqrt{3})$$

$$\frac{1}{2}x^2 + x < 1$$

$$x^2 + 2x < 2$$

$$x^2 + 2x - 2 < 0$$

$$m \quad x^2 - 2x + 4 > 0$$

$$\Delta = \sqrt{4 - 4 \cdot 3 \cdot 4}$$

$$\Delta = \sqrt{-80} \quad \text{no } 2x +$$

$$o \quad x^2 + 2x > 7$$

$$x^2 + 2x - 7 > 0$$

$$\Delta = \sqrt{2^2 - 4 \cdot 1 \cdot (-7)} = \sqrt{4 + 28} = \sqrt{32} = 4\sqrt{2}$$

$$x = \frac{-2 \pm 4\sqrt{2}}{2}$$

$$\Delta = 4\sqrt{2}$$

$$x = \frac{-2 \pm 4\sqrt{2}}{2} \quad \left(\begin{array}{l} -1+2\sqrt{2} \\ -1-2\sqrt{2} \end{array} \right) \quad (-\infty, -1-2\sqrt{2}) \cup (-1+2\sqrt{2}, +\infty)$$